```
r1.high, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood
     1 20263016             5000           0.278              -35631.6
     2 20263017             5000           0.277              -35646.9
     3 20263018             5000           0.276              -35625.8
     4 20263019             5000           0.274              -35657.4
 best.log.likelihood errored.proposals
            -35604.5                 0
            -35643.7                 0
            -35622.8                 0
            -35638.4                 0
                                    file
 mcmc_r1.high_step2_chain1_2026-10-19.Rdata
 mcmc_r1.high_step2_chain2_2026-10-20.Rdata
 mcmc_r1.high_step2_chain3_2026-10-20.Rdata
 mcmc_r1.high_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
              hiv.specific.mortality.rates.2
                                       2.748
                                     trate.3
                                       2.685
      age.30.to.39.transmission.multiplier.3
                                       2.296
      age.30.to.39.transmission.multiplier.1
                                       2.182
                    female.to.male.multiplier
                                       2.175
              hiv.specific.mortality.rates.1
                                       2.134
                                     trate.1
                                       2.114
            male.hiv.mortality.multiplier.2
                                       1.921
   age.50.and.over.transmission.multiplier.3
                                       1.918
      age.15.to.19.transmission.multiplier.3
                                       1.863
      age.0.to.14.hiv.mortality.multiplier.1
                                       1.744
      age.20.to.29.transmission.multiplier.1
                                       1.687
            male.hiv.mortality.multiplier.1
                                       1.641
      age.0.to.14.hiv.mortality.multiplier.2
                                       1.613
      age.20.to.29.transmission.multiplier.3
                                       1.587
     age.15.to.24.hiv.mortality.multiplier.1
                                       1.525
     age.15.to.24.hiv.mortality.multiplier.2
                                       1.510
age.65.to.79.mortality.slope.multiplier.male
                                       1.507
                                     trate.4
```

r1.high step 2: likelihood

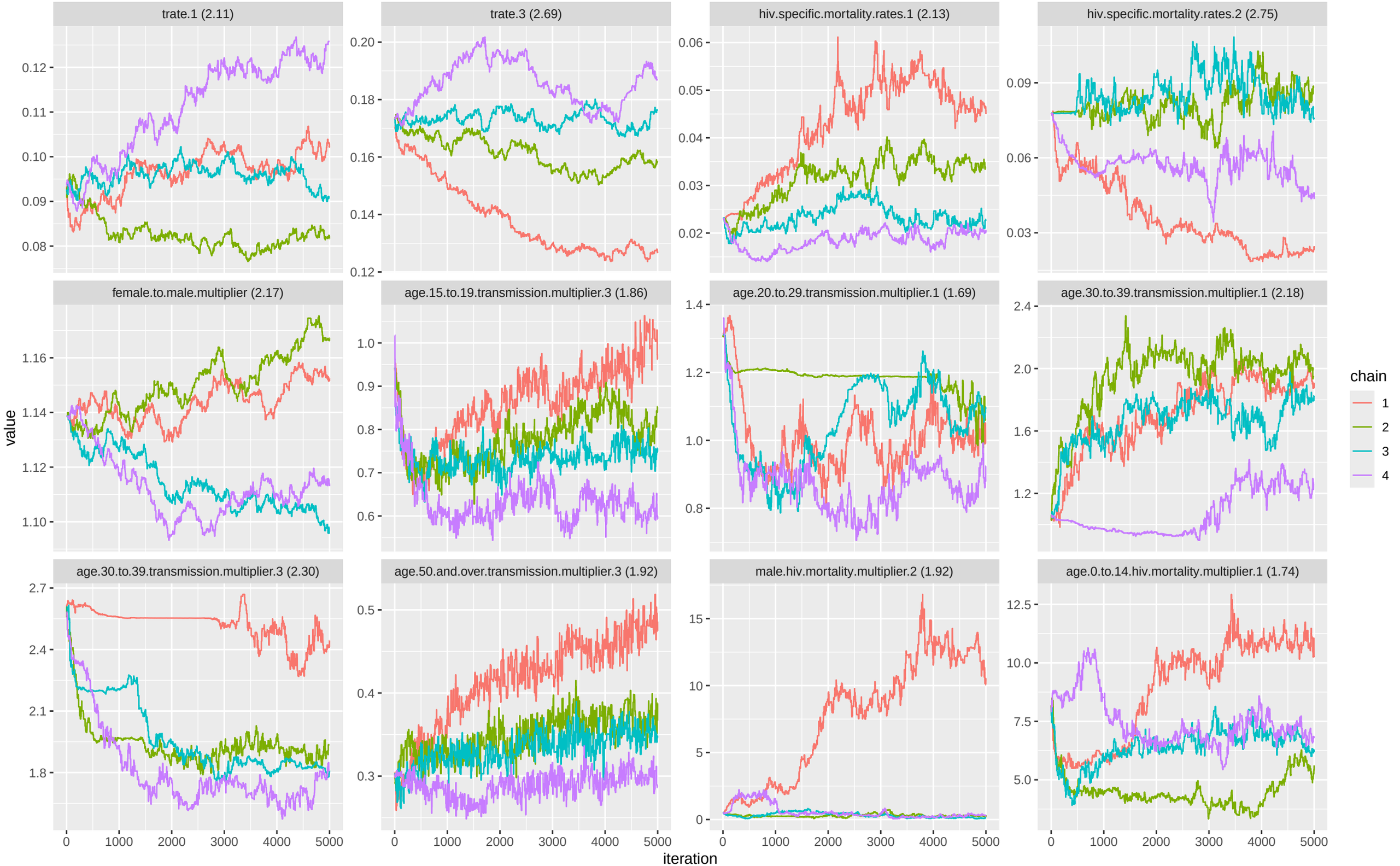
0
−10000
−20000
−30000
lue

0
25000
50000
75000
100000
iteration

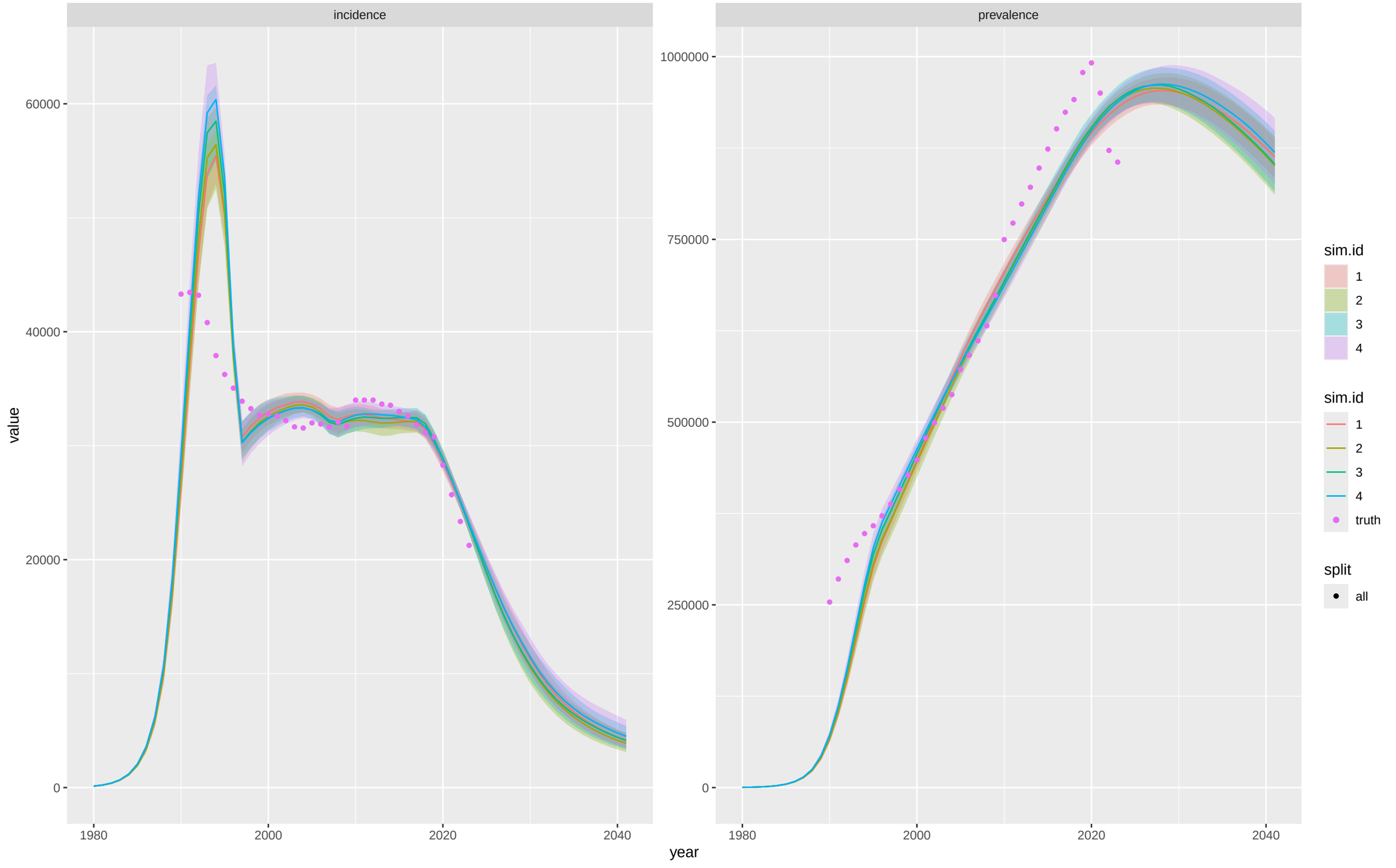
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

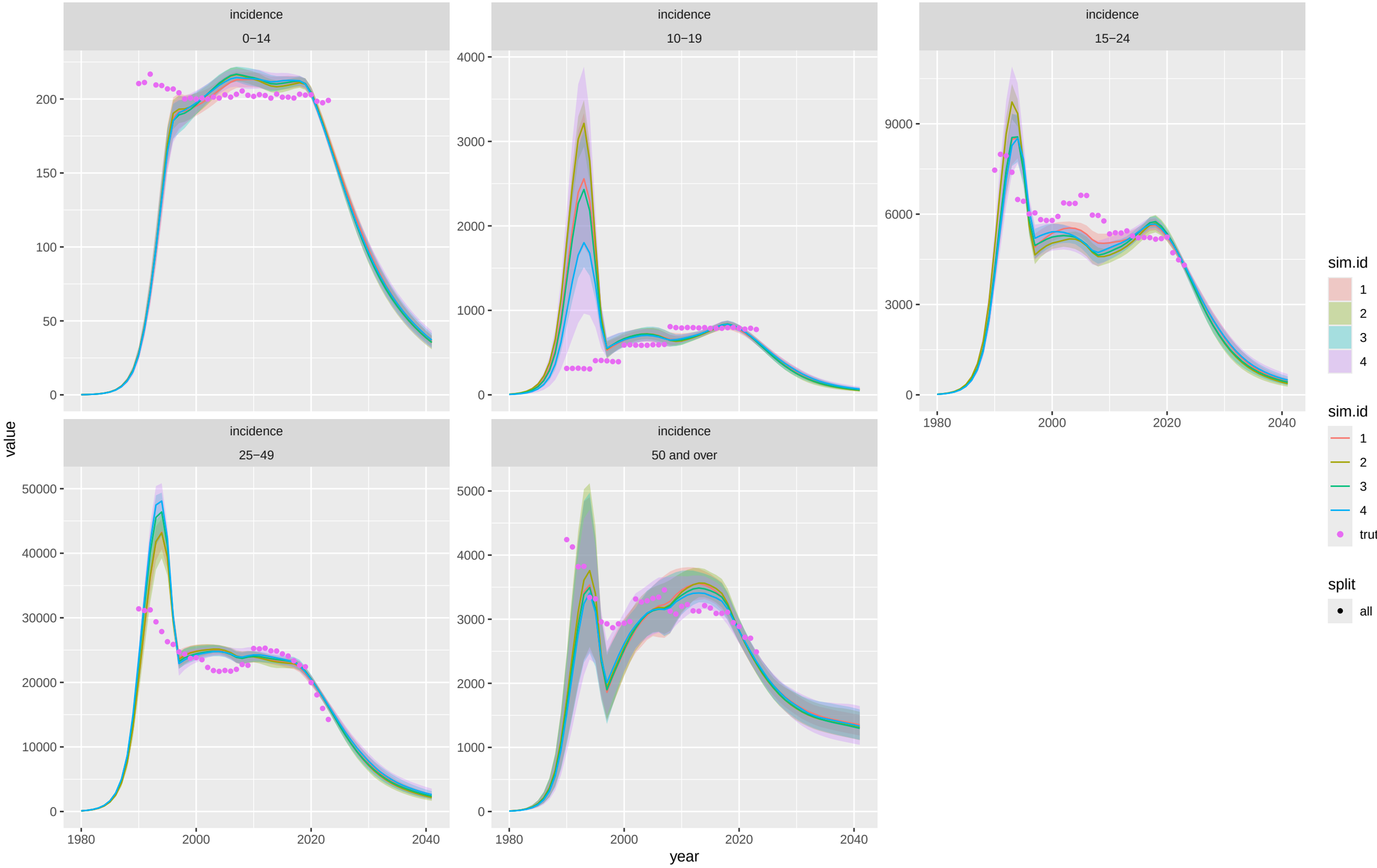
r1.high step 2: trace



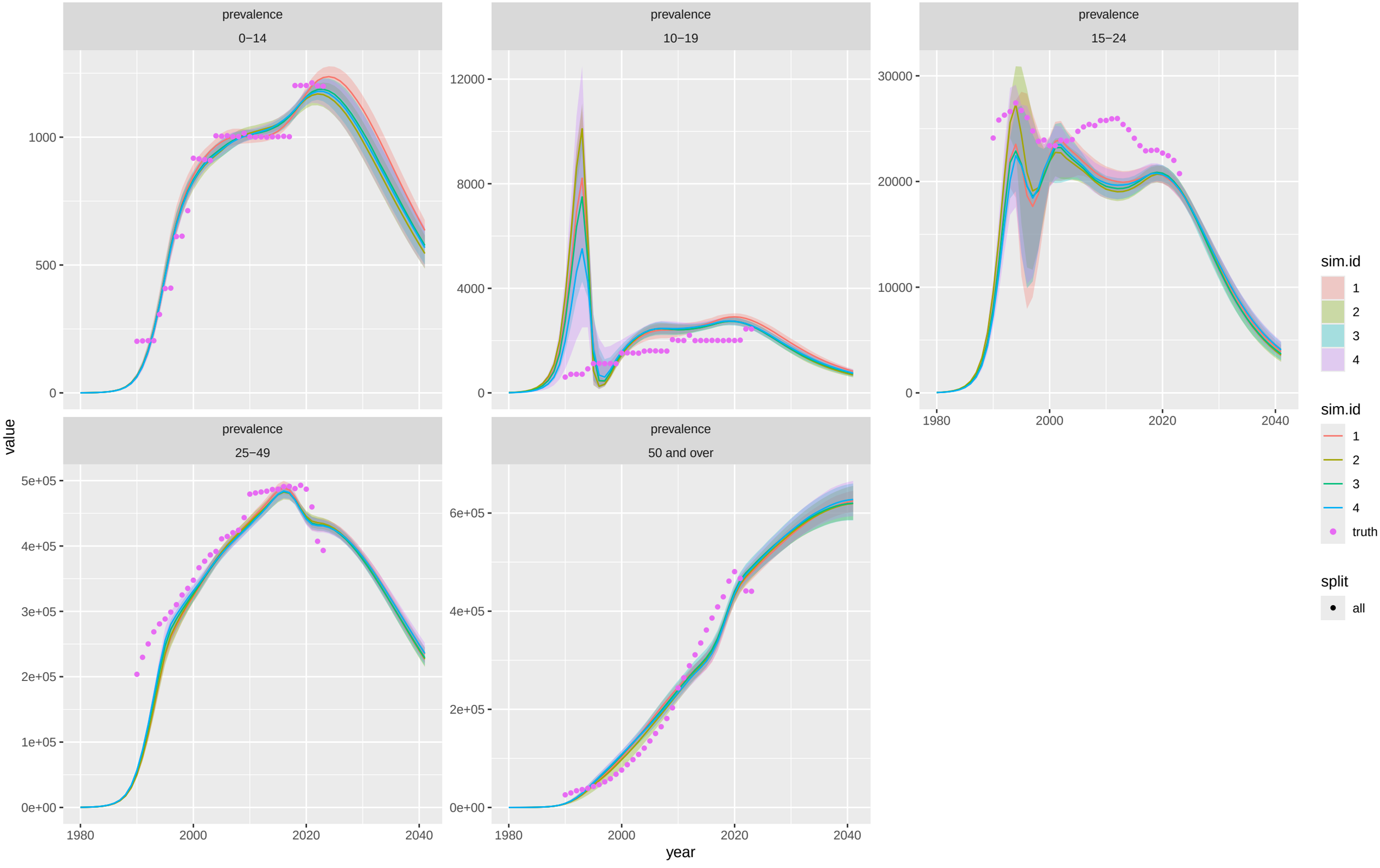
r1.high step 2: fit (one color per chain)



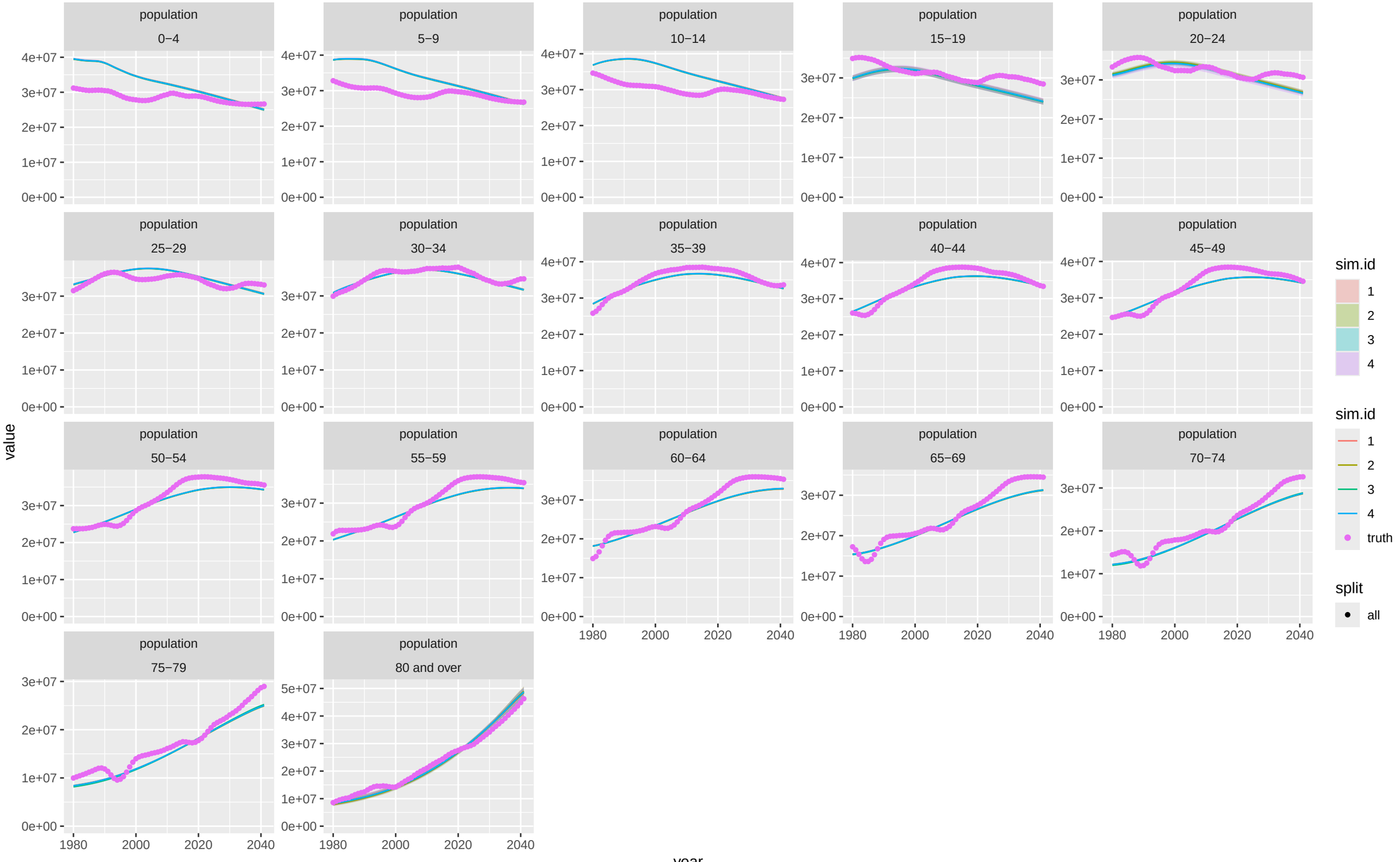
r1.high step 2: incidence.by.age (one color per chain)



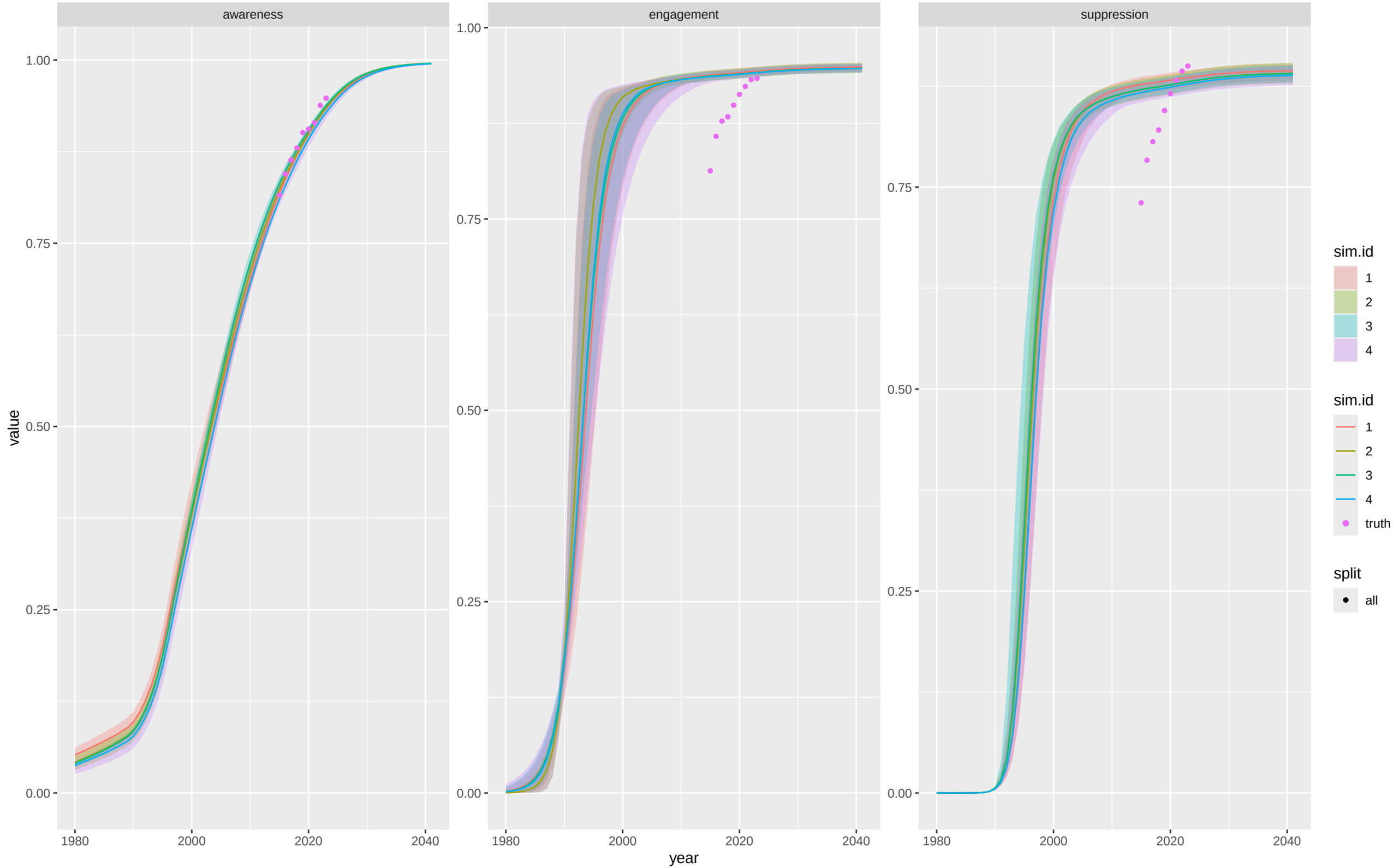
r1.high step 2: prevalence.by.age (one color per chain)



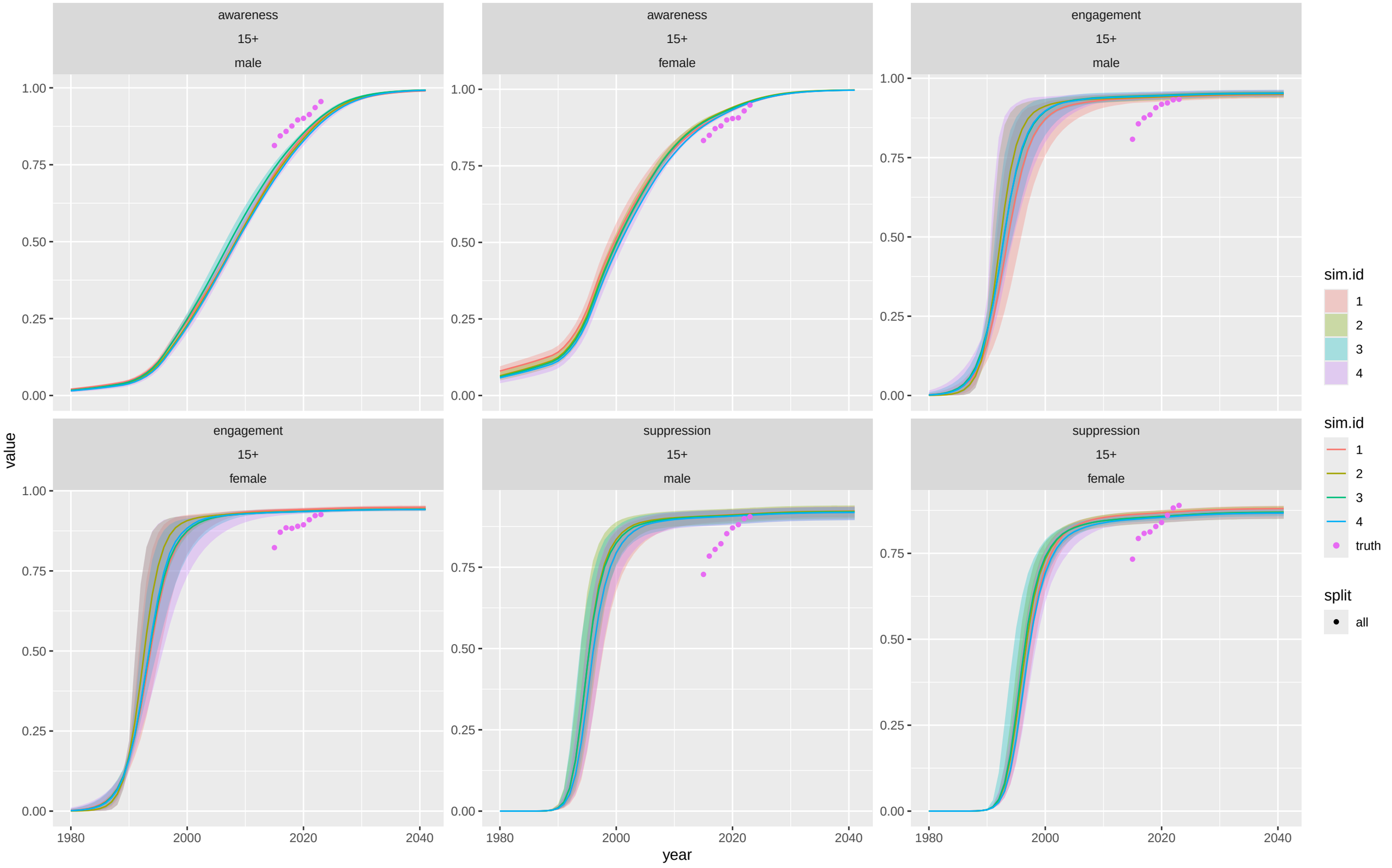
r1.high step 2: population.by.age (one color per chain)



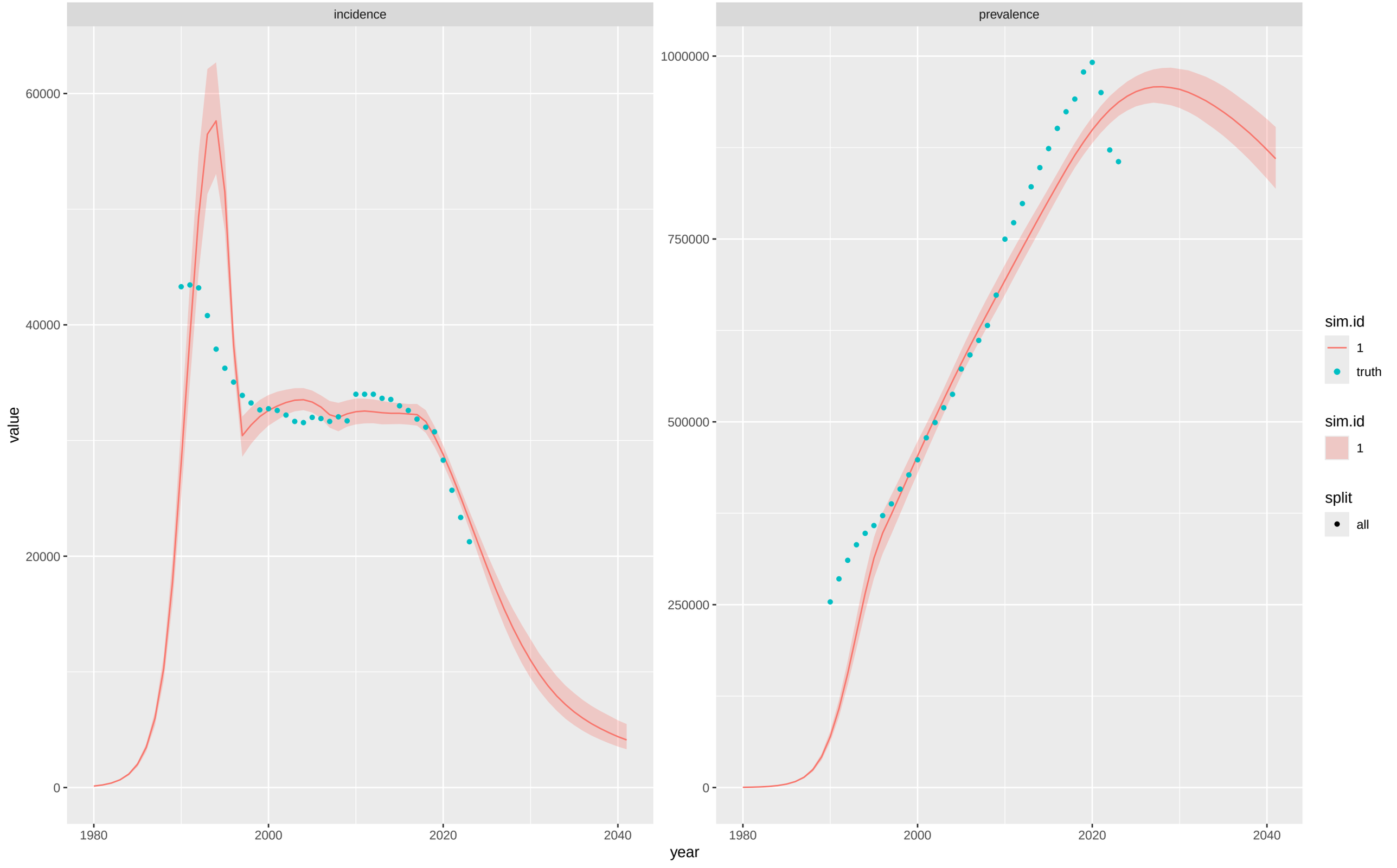
r1.high step 2: cascade (one color per chain)



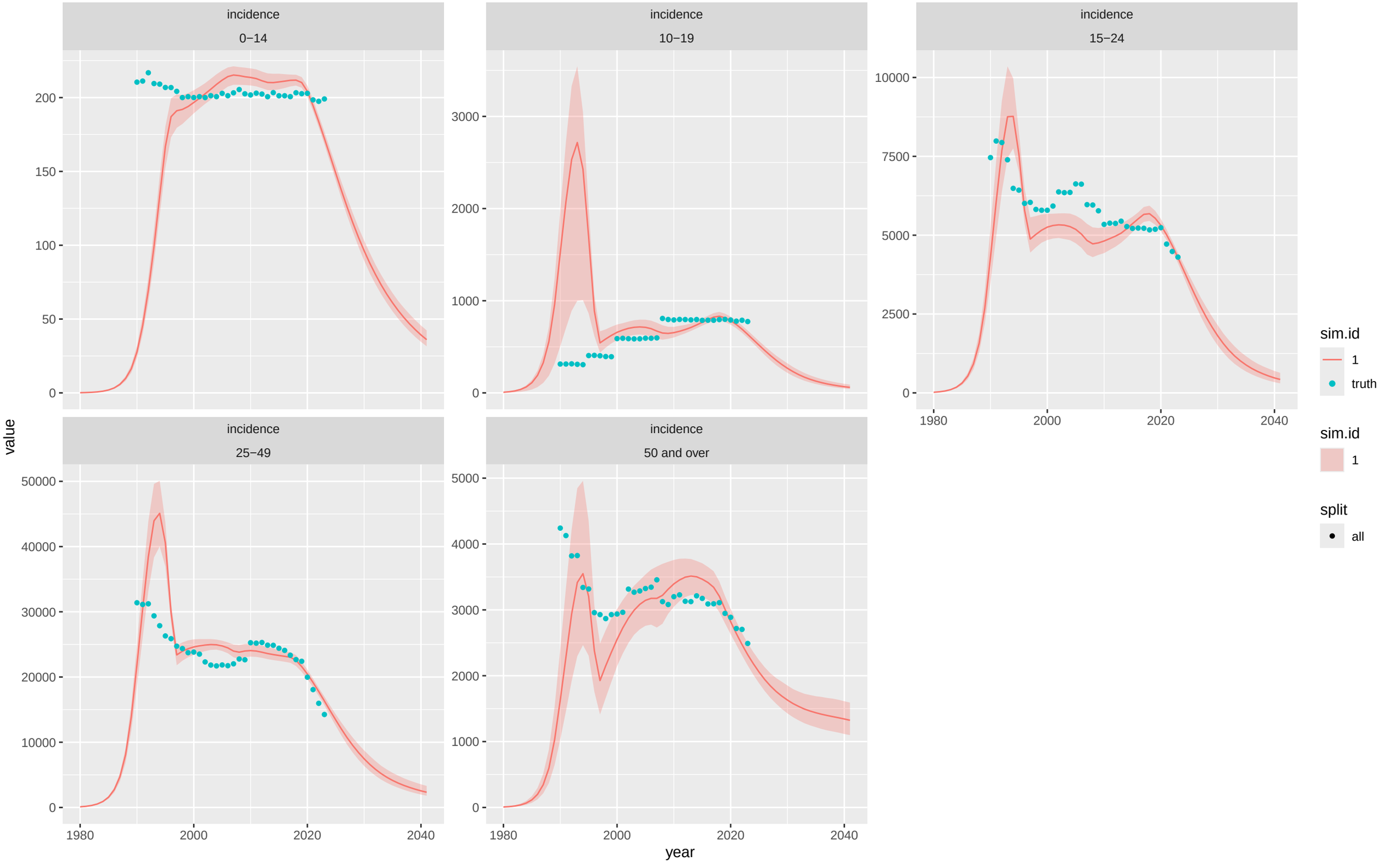
r1.high step 2: cascade.by.age.sex (one color per chain)



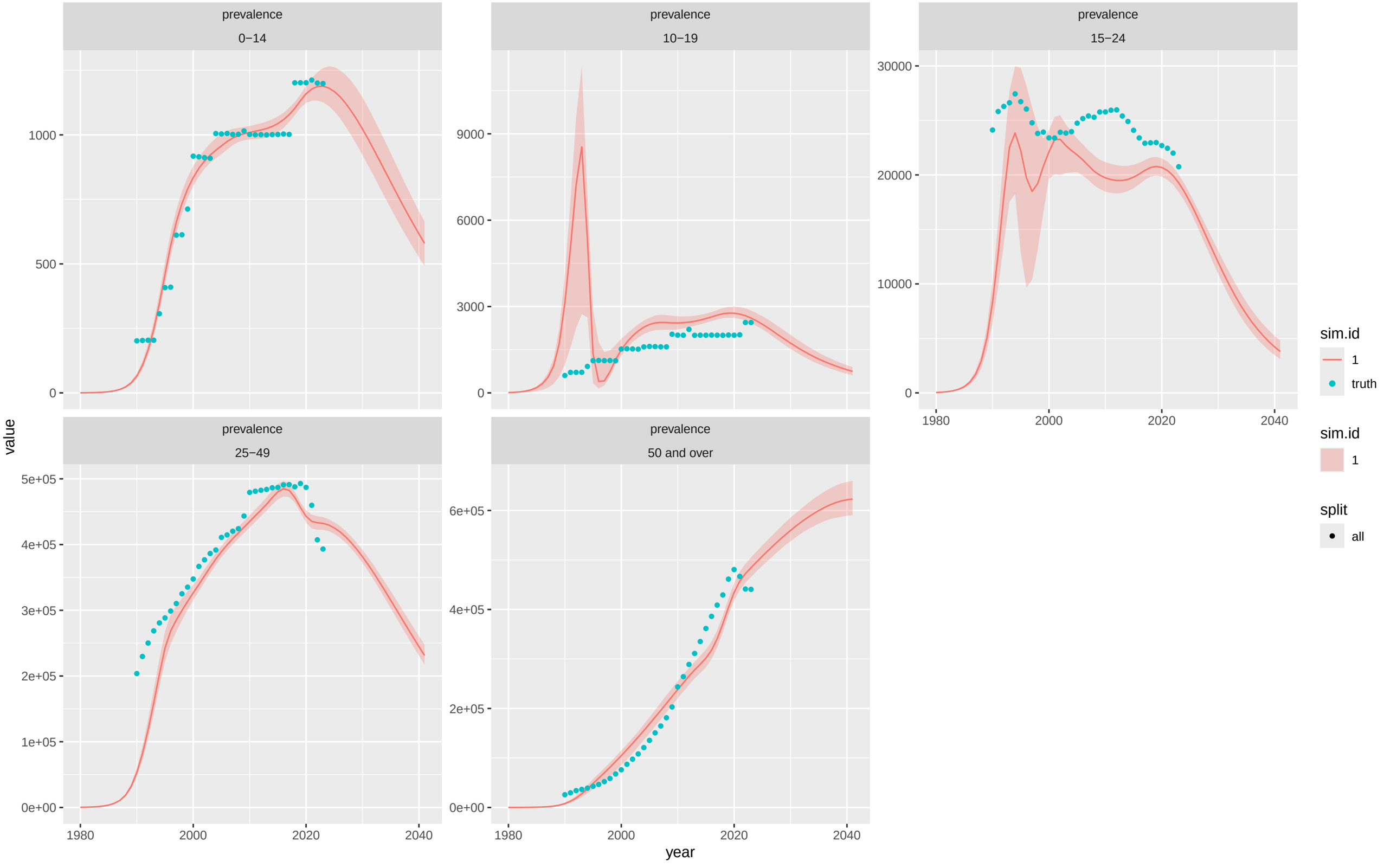
r1.high: fit (merged Step 2 simset)



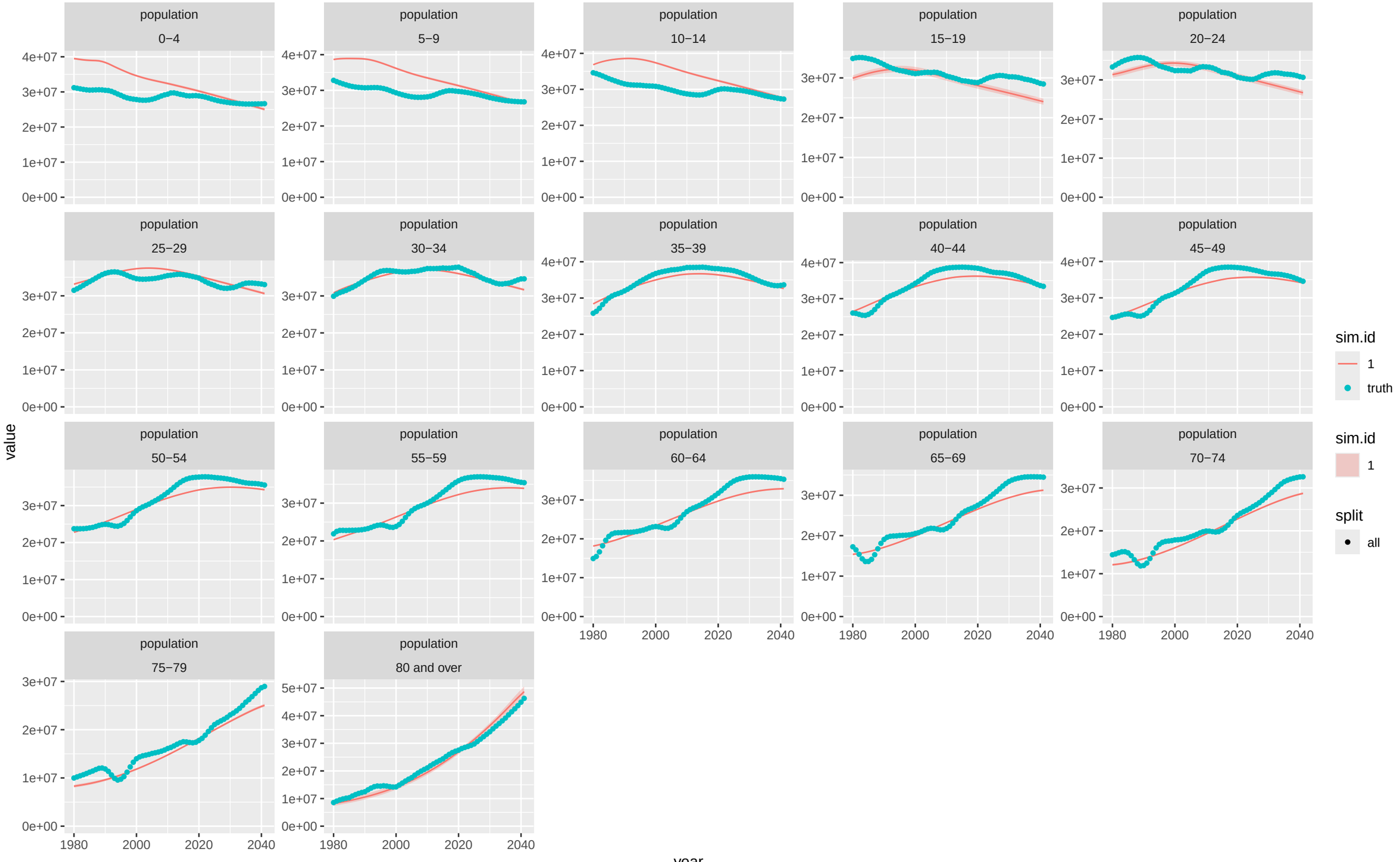
r1.high: incidence.by.age (merged Step 2 simset)



r1.high: prevalence.by.age (merged Step 2 simset)



r1.high: population.by.age (merged Step 2 simset)



r1.high: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

1980
2000
2020
2040

sim.id
1
truth

sim.id
1

split
all

r1.high: cascade.by.age.sex (merged Step 2 simset)

